

# An IoT-Based Vehicle Accident Detection and Classification System Using Sensor Fusion

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**Abstract**—Road accidents are a leading cause of death and disability among youth. Contemporary research on accident detection systems is focused on either decreasing the reporting time or improving the accuracy of accident detection. Internet-of-Things (IoT) platforms have been utilized considerably in recent times to reduce the time required for rescue after an accident. This work presents an IoT-based automotive accident detection and classification (ADC) system, which uses the fusion of smartphone's built-in and connected sensors not only to detect but also to report the type of accident. This novel technique improves the rescue efficacy of various emergency services, such as emergency medical services (EMSs), fire stations, towing services, etc., as knowledge about the type of accident is extremely valuable in planning and executing rescue and relief operations. The emergency assistance providers can better equip themselves according to the situation after making an inference about the injuries sustained by the victims and the damage to the vehicle. In this work, three machine learning models based on Naïve Bayes (NB), Gaussian mixture model (GMM), and decision tree (DT) techniques are compared to identify the best ADC model. Five physical parameters related to vehicle movement, i.e., speed, absolute linear acceleration (ALA), change-in-altitude, pitch, and roll, have been used to train and test each candidate ADC model to identify the correct class of accident among collision, rollover, falloff, and no accident. NB-based ADC model is found to be highly accurate with 0.95 mean F1-score.

**Index Terms**—Accident detection and classification (ADC), decision tree (DT), Gaussian mixture model (GMM), Internet of Things (IoT), Naïve Bayes (NB), sensor fusion.

## I. INTRODUCTION

DEATHS and disabilities by road accidents are increasing with each passing year. An increase in population and per capita incomes has led to an increase in ownership and the presence of vehicles on roads. Greater traffic volumes, over-speeding, reckless and drunken driving, driver fatigue, poor road infrastructure, and the presence of animals on roads are some reasons that are responsible for road fatalities. According to the World Health Organization (WHO), the percentage of road accident fatalities to the total number of deaths worldwide has increased by 2.2% [1]. Approximately 1.35 million people die due to road accidents every year [2].

Manuscript received May 10, 2020; revised June 16, 2020; accepted June 30, 2020. Date of publication July 13, 2020; date of current version January 7, 2021. (Corresponding author: Nikhil Kumar.)

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Digital Object Identifier 10.1109/IIOT.2020.3008896

In many cases, human lives are lost in road accidents due to delays in emergency medical assistance. According to Golden Hour Principle [3], there is a high probability that timely medical and surgical aid can avoid death during the golden hour, which is the period after the traumatic injury. A decrease in the response time of emergency medical care can reduce the probability of death by one-third on an average [4]. The percentage of people who die before reaching the hospital in low- and middle-income countries is more than twice as compared to high-income countries [2]. In the recent past, information and communication technology, such as Internet of Things (IoT) has been used to decrease the accident rescue time. IoT is an interconnection of a vast variety of embedded and smart devices, such as computers, smartphones, smart sensors and actuators, embedded processors, etc., with the modern Internet. IoT is a potential medium for tracking and control of smart automobiles that can link any connected physical unit to a control server [5]. Most researchers have confined their work to improving the accuracy of accident detection, estimating the severity of road accidents or minimizing the rescue time post-occurrence of an accident [6], [7]. In addition, most systems to detect and report road accidents are expensive and limited to high-end vehicles. Another drawback of the current systems is their inability in identifying the type of accident as a collision, rollover, or a falloff event. Merely reporting the occurrence of an accident event severely limits the ability of the emergency rescue workers to provide the victims with the right kind of rescue support and medical aid. The information about the type of accident sustained provides valuable information about the damages sustained by the vehicle and the extent of injuries sustained by the victims.

This research work answers the following questions.

- 1) Can there be an inexpensive accident detection and classification (ADC) system that can be retrofitted to any vehicle?
- 2) Which is the best suited machine-learning classification model that can accurately detect and classify the road accident type?
- 3) How to automatically report the occurrence of road accident with its type and location to the relevant agencies by sending the emergency notification if the accident victims are incapacitated?

**Contribution of This Work:** Research work on developing the mechanisms for prediction, prevention, detection, and management of road accidents are predominantly focused on either enhancing precision or reducing rescue time following the occurrence of road accidents. In this study, we have proposed

a smartphone-based end-to-end IoT system architecture to detect and notify automotive accidents. The work also focuses on identifying the best machine-learning ADC model based on either of three different classifiers, namely, Gaussian mixture model (GMM), decision tree (DT), and Naïve Bayes (NB). This classification can be extremely helpful in the planning and execution of rescue operations. The proposed solution is inexpensive and efficient that can be easily retrofitted in any class and model of the vehicle. Experiments and analysis of the results have shown that in all proposed solutions NB-based model is highly accurate in the detection and classification of road accidents.

This article is organized as follows. Major related work in the area of automatic vehicle accident detection and multi-sensor fusion is described in Section II. The architecture of the proposed IoT system, workflow diagram of the system, model variables, and preprocessing methods are explained in Section III. The hardware and software setup of the system is proposed in Section IV. Next, Section V describes the possible accident scenarios, and then there is a description of the experimental setup and data collection in Section VI. Three different ADC models, based on GMM, DT, and NB classifiers, are proposed in Section VII. Results, comparative analysis of proposed models, and discussion are presented in Section VIII while conclusion and future work are laid out in Section IX.

## II. RELATED WORK

Research work related to the detection, localization, reporting, modeling, and analysis of road accidents is discussed in this section. Several authors have reported smartphone-based accident detection systems. Aloul *et al.* [8] developed an Android application that uses the accelerometer data to develop an accident detection model based on dynamic time warping (DTW) and hidden Markov models (HMMs). An SMS, which includes accident location and severity, is sent to the emergency medical service (EMS), police, and family on detection of an accident. Bhatti *et al.* [9] have used built-in speed, location, pressure, sound, and g-force sensors of a smartphone to develop a low cost, portable solution that detects and reports an accident to the nearest hospital.

Shaik *et al.* [10] proposed an IoT-based vehicle collision detection system that uses a ADXL345 accelerometer to detect the collision and GPS to detect and send the accident location to the nearest ambulance via Internet. Sharma and Sebastian [11] proposed an algorithm for vehicle accident detection that uses ADXL345 accelerometer and vibration sensor to detect the vehicle crash, and heart rate sensor to monitor health of the passenger, and send an emergency SMS or voice call to the family. An accelerometer and an ultrasonic sensor are used by Priya *et al.* [12] in an AtMega162-microcontroller-based accident detection system to notify the nearest first aid center about the accident location via SMS.

To reduce delay in reporting a road accident, Dar *et al.* [13] have proposed a fog computing-based approach to build a low-cost smartphone-based system. The system uses an Android application that takes inputs from smartphone sensors to detect an accident. The system locates and notifies a nearby hospital

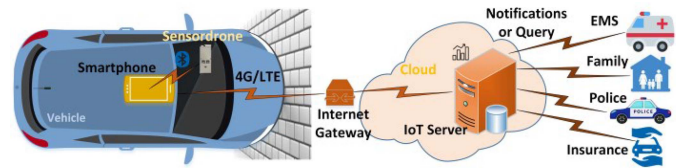


Fig. 1. Architecture of the proposed IoT system.

using GPS when an accident is detected. Fernandes *et al.* [14] have developed an Android application, which receives inputs from the smartphone sensors to detect collisions and rollovers, as well as receive road hazard warnings issued by vehicles in the vicinity.

When a good Internet connection is limited to some nearby vehicles, intervehicle communication approaches, such as vehicular *ad hoc* network (VANET) and Internet of Vehicles (IoV), may help vehicles carry out reporting of accidents via each other. Fabian *et al.* [15] have introduced a programmable objective function (POF) and developed an architecture inspired from a software-defined network (SDN) approach to improve the Quality of Service (QoS) in dynamic IoV networks, which can be very important while vehicle reporting the vehicle accidents to each other. Rachedi and Badis [16] have proposed a programmable and hybrid architecture based on the virtual backbone (VB) formed by selected parked vehicles. Architecture is used to reduce the use of cellular communication link in IoV, which is based on connected dominated sets (CDSs) and software-defined vehicular networks (SDVNs) approaches. Mekki *et al.* [17] have discussed a vehicular cloud access (VCA) problem that is modeled as an evolutionary game, and proposed two VCA algorithms based on the evolutionary game (EG-VCA), and distributed *Q*-learning (QL-VCA).

Acharya *et al.* [18] and Ibrahim *et al.* [19] have developed systems that used accelerometer and gyroscope data to detect and report vehicle rollover events. Smolka and Skublewska-Paszkowska [20] presented a smartphone-based collision detection system using accelerometer, magnetometer, and GPS module to reduce the number of false alarms. Sadeky *et al.* [21] have used real-time video footage of traffic surveillance systems to construct the histogram of flow gradient (HFG). The HFG features are used as inputs to develop the statistical logistic regression model to predict road accidents. None of these systems reports the ability to determine the type of accident, which can be useful in planning the rescue type and estimating the number of emergency services required.

The fusion of sensor data for detection and analysis of road accidents is not reported in any major work, which can help to improve the ADC performance. Felisberto *et al.* [22] presented a system, which used the fusion of data from a network of wireless sensors to detect accidents in elderly people. The authors claimed that the proposed solution is nearly 100% accurate in detecting normal falls. Damousis and Tzovaras [23] have used a fuzzy expert system for the fusion of the most significant features extracted from the eyelid activities of vehicle drivers. The system is found to be efficient in predicting sleep onset and accidents. Zhang *et al.* [24] have attempted the

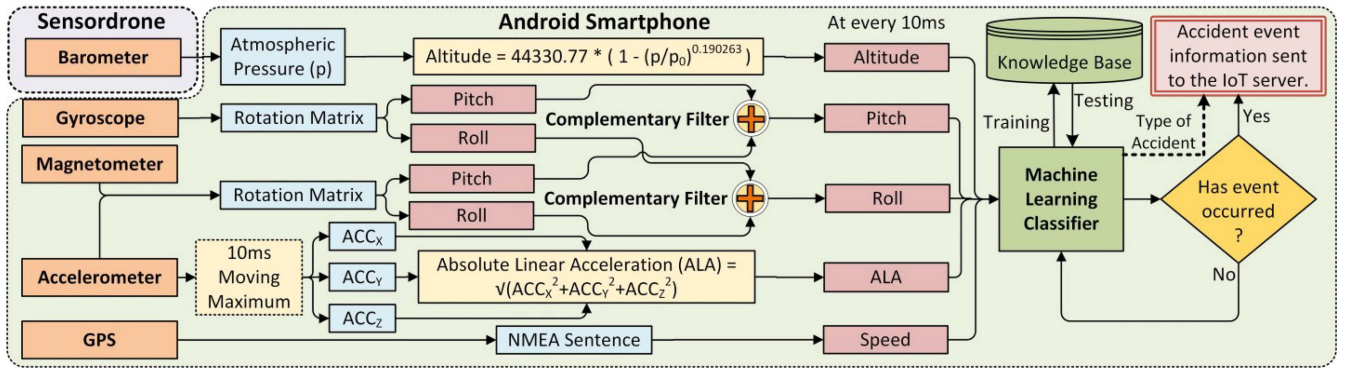


Fig. 2. Workflow diagram of the accident classification system.

fusion of multisource data by combining traffic metrics and social media tweets for real-time detection of road accidents. Using support vector machines (SVMs) as the classification model and implementing fivefold cross-validation, the authors claim an increase in prediction accuracy by the integration of these two data sources.

Moulik and Majumdar [25] have developed an IoT system that uses the fusion of multiple ultrasonic sensors. The system uses infrared transmitter–receiver pairs and a fuzzy inference system to detect the accidental fall of humans. The multisensor fusion is claimed to achieve an improvement of 16% as compared with existing approaches.

Although, previous works have proposed some systems for collision/rollover detection and notification, to the best of our knowledge there is no system that classifies these accidents as collision, falloff, rollover, and no accident. This work presents a novel smartphone sensor fusion-based solution for detection and categorization of accident.

### III. ARCHITECTURE OF THE SYSTEM

#### A. Architecture of Our Proposed IoT System

An IoT architecture is proposed in Fig. 1 to address the problem of vehicle accident classification. We have used a modern smartphone and a prefabricated sensor platform termed as Sensordrone [26] to obtain the values of different physical parameters relevant to vehicle motion. The Android smartphone contains several integrated sensors, such as accelerometer, gyroscope, magnetometer, GPS, etc., which can be used to determine the speed, direction, rotation, and g-force, etc., of the vehicle. The Sensordrone remains attached to the smartphone via Bluetooth connection to send the data to the smartphone. Most of the processing is handled by the smartphone. Processing in smartphone substantially decreases the Internet resource (e.g., bandwidth, etc.) usage by transmitting only relevant details, such as venue, name, nature of the accident, etc., to the IoT server. The IoT server delivers emergency alerts to various emergency services, such as EMS, local police station, and fire service, and other receivers, such as relatives, insurers, blood donor, towing service, etc., after assessing the situation.

#### B. Operating Workflow Diagram

The operating flow diagram of the proposed ADC system is depicted in Fig. 2. The feature vector of our classification

model includes five parameters, namely, speed, absolute linear acceleration (ALA), change-in-altitude, pitch, and roll, to detect and classify a vehicle accident. The system is trained and a knowledge base having four accident classes is created by providing the training-data set to the machine learning classifier. After training, the system model is tested against the testing data set; and then accuracy is measured to ascertain the delivery of the valid notification to the emergency services, in case of an accident.

#### C. Model Variables

1) *Speed*: It is quite logical that when a car experiences a serious accident, whether it is a collision, a rollover or a falloff, its speed would become zero ultimately. Smartphone's GPS has been used to measure the speed of the vehicle. Each GPS device receives National Marine Electronics Association (NMEA) sentence from satellites containing information related to the location, velocity, and time of the device [27]. Each class of GPS devices does have its own NMEA sentence. The most common NMEA sentence that is used by most of the Android devices is \$GPRMC. Following is an example of the \$GPRMC NMEA sentence, where the velocity is 018.8 knots:

```
$GPRMC,152844,A,1831.335,N,07734.331,E
018.8,054.4,030220,003.1,W*6A.
```

2) *Absolute Linear Acceleration*: Whenever a running vehicle drop from a height, the orientation of the vehicle will not be the same as it was on the road before falling. In such a case, the vehicle cannot remain parallel to the gravitational axes ( $X$ ,  $Y$ , and  $Z$ ) and deceleration due to the impact that can be distributed in more than one axis. Therefore, it is very difficult to assess the precise value of the deceleration when it is distributed in  $X$ ,  $Y$ , and  $Z$  components. To address this problem, ALA (or signal magnitude vector [28]) is calculated with the help of  $X$ ,  $Y$ , and  $Z$  components of deceleration. ALA is independent of the vehicle's orientation during falloff or collision directions (whether the vehicle is moving or steady). ALA shows the acceleration characteristic parallel to the direction of impact. This absolute quantity is a resultant vector of  $X$ ,  $Y$ , and  $Z$  components of deceleration. ALA is determined as

$$ALA = \sqrt{(DEC_X)^2 + (DEC_Y)^2 + (DEC_Z)^2} \quad (1)$$

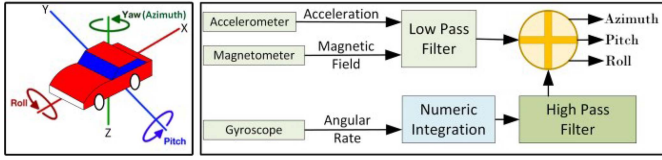


Fig. 3. (Left) yaw, pitch, and roll; (right) complementary filter.

where  $DEC_X$ ,  $DEC_Y$ , and  $DEC_Z$  are the decelerations at  $x$ -axis,  $y$ -axis, and  $z$ -axis, respectively. Smartphone's accelerometer has been used to measure the ALA whose measurement unit is  $g$  ( $\approx 9.80665 \text{ m/s}^2$ ). It is mentioned in the literature that when a vehicle collided with an object (or any solid surface) at the speed of 23 km/h or more, the intensity of deceleration always cross  $5g$  [29].

3) *Change-in-Altitude in 1 s*: Knowledge of altitude shift enables the system to predict the falloff of the vehicle. During the falloff, the altitude of the vehicle is the most accountable feature that shifts drastically. Although GPS can also be used to determine the altitude, GPS is not acceptable when the position stays the same while the altitude shifts. No specialized instrument is present that can determine altitude solely, but it can be determined using atmospheric pressure  $P$  and temperature. In this experiment, the altitude is calculated by using the pressure  $P$  measured by the Sensordrone's barometric altimeter. The atmospheric temperature is neglected because it does not substantially change at same coordinates. If  $P_0$  is reference pressure recorded at the sea level, final formula of calculating the altitude would come out as provided by Ye *et al.* [30]

$$\text{Altitude} = 44330.77 * \left(1 - (P/P_0)^{0.190263}\right). \quad (2)$$

4) *Roll and Pitch*: The rotation angles of an object around the  $x$ - and  $y$ -axis are known as roll and pitch, respectively. Roll and pitch are calculated by the fusion of rotation information provided by the gyroscope, accelerometer, and magnetometer using a complementary filter. The rotation matrix rotates the Euclidean space counterclockwise around the center of the Cartesian coordinate system by angle  $\theta$ . We found that if either roll or pitch is greater than  $90^\circ$ , a rollover happened. The longitudinal axis is always supposed to be  $x$ -axis and the lateral axis is always supposed to be  $y$ -axis [Fig. 3(left)].

#### D. Preprocessing Methods

1) *Sensor Fusion Using Complementary Filter*: The direction and orientation of the vehicle can be determined by pitch, roll, and azimuth (yaw). The longitudinal, lateral, and vertical axes of the vehicle's body are expressed by  $X$ ,  $Y$ , and  $Z$ , respectively. Roll, pitch, and azimuth (yaw) are the rotating angles around the  $x$ -,  $y$ -, and  $z$ -axis, respectively [Fig. 3(left)]. Three inertial sensors, *viz.*, gyroscope, accelerometer, and magnetometer have been used to determine the orientation of the vehicle.

Despite their respective properties, owing to their shortcomings and drift in measurements, no stand-alone inertial sensor is adequate to calculate the orientation effectively. The magnetometer can measure azimuth precisely, but not pitch and roll, whereas accelerometer can calculate pitch and roll

precisely, but not azimuth. Collectively, all sensors can calculate rotational speed and rotational angle, about all three axes. At high frequency, short-term estimates of the combination of the accelerometer and magnetometer are not accurate owing to a few small external factors, except the  $g$ -force. A low-pass filter (only makes low-frequency signals from 0 Hz to its threshold level) is required for correction because the combination of both of these inertial sensors works well at low frequency.

Another sensor that can calculate the angular momentum about the  $x$ -,  $y$ - and  $z$ -axis is the gyroscope, but it is also exposed to strong drift. Gyroscope operates well at a high rate, so high-pass filtering (allows only high-frequency signals from its threshold frequency) is required to fix the results because the measurements start drifting in long run.

Each sensor has its pros and cons, but their fusion can improve the performance anyway [31]. In this research, a complementary filter [32] is used, consisting of both a low-pass filter and a high-pass filter [Fig. 3(right)]. The complementary filter provides data that are more precise by applying the weight fractions to the outputs of low-pass and high-pass filters. If  $A_{gyro}$ ,  $P_{gyro}$ , and  $R_{gyro}$  are the corresponding azimuth, pitch, and roll provided by the high-pass filter, and  $A_{acc\_mag}$ ,  $P_{acc\_mag}$ , and  $R_{acc\_mag}$  are the corresponding azimuth, pitch, and roll provided by the low-pass filter, then the resultant azimuth, pitch, and roll determined by the complementary filter are as follows:

$$\text{Azimuth} = \alpha * A_{gyro} + (1 - \alpha) * A_{acc\_mag} \quad (3)$$

$$\text{Pitch} = \alpha * P_{gyro} + (1 - \alpha) * P_{acc\_mag} \quad (4)$$

$$\text{Roll} = \alpha * R_{gyro} + (1 - \alpha) * R_{acc\_mag}. \quad (5)$$

The gyroscope gives more stable value than accelerometer and magnetometer collectively, and weightage of gyroscope must be near 0.95 while using complementary filter [33]. We have tried to calibrate the combination for different values of  $\alpha$  and find that this combination was showing smoother curve of pitch, roll, and azimuth at  $\alpha = 0.98$ .

2) *Maximum Deceleration Using 10-ms Moving Maximum*: The standard signal generation rate of an accelerometer might be more than 2000 Hz. It is quite challenging to process the data and report the maximum ALA value at such a high frequency, especially when we need accurate decisions and our device has limited resources and processing capabilities. Iyoda *et al.* [34] have used a 10-ms moving average technique to deal with this problem; however, the moving average can diminish the ALA's peak value. To attain the highest possible ALA, we are using a 10-ms Moving Maximum method that registers the peak value for each and every 10-ms time window as  $\dots, S - 2, S - 1, S, S + 1, \dots$  (Fig. 4).

#### IV. HARDWARE AND SOFTWARE SETUP OF THE SYSTEM

##### A. Hardware Setup

In this work, a complete hardware setup comprises a *SAMSUNG Galaxy S8* Android smartphone, a *Sensordrone*, and a 1:12 scaled RC car. The hardware setup is depicted in the inset figure of Fig. 6.

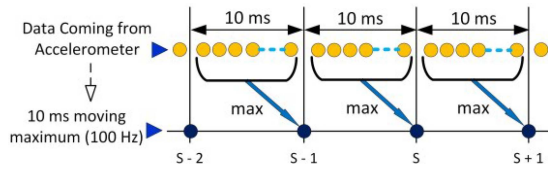


Fig. 4. 10-ms moving maximum.

TABLE I  
RC CAR SPECIFICATIONS

| Vehicle Parameter     | Value                       |
|-----------------------|-----------------------------|
| Brand Name            | GPTOYS Foxx S911 RC Car     |
| Dimension (L * W * H) | 31.0 cm * 26.5 cm * 15.0 cm |
| Weight                | 1.078 Kg                    |
| Scale Ratio           | 1:12                        |
| Track Width           | 22.0 cm                     |
| Wheelbase             | 20.7 cm                     |
| Ground Clearance      | 40.0 cm                     |
| Maximum Speed         | 53 km/h                     |
| Transmitter Frequency | 2.4 GHz 4Ch                 |
| Operating Distance    | 80 m                        |
| Maximum Turning Angle | 45 degrees                  |

1) *Samsung Galaxy S8 Android Smartphone*: In this work, besides of external sensors, the *SAMSUNG Galaxy S8* Android smartphone's inbuilt 9-axis inertial sensor and GPS sensor have been used. Inertial sensor, whose measuring range is  $\pm 179^\circ$  and  $\pm 16g$ , is used to measure the pitch, roll, and deceleration, and a GPS sensor is used to determine the vehicle's speed and location. 1.9-GHz clock speed of *Exynos 8895* microprocessor and 4 GB of RAM makes it a suitable computing device for our experiments.

2) *Sensordrone*: *Sensordrone* is a tiny sensor hub that includes seven separate in-house sensors to assess twelve different environmental variables, such as temperature, humidity, CO, atmospheric pressure, light, proximity, etc., [26]. *Sensordrone* is a programmable device, which has a UART port to attach an additional sensor (e.g., CO<sub>2</sub> sensor) and Bluetooth to establish a connection with any processing device. In this research, only barometric altimeter sensor (measurement range: 26kPa to 126kPa) of *Sensordrone* is used to determine the altitude of the vehicle.

3) *RC Car*: A 1:12 scaled toy Radio Controlled (RC) Car [35] is used to mimic real-life accident scenarios since conducting the accident experiments with an actual car is neither viable nor budget effective. This RC car is made of heavy-performance ABS material and has metallic shock absorbers. The configuration of RC car is specified in Table I.

## B. Software Setup

Two Android applications, *SNUSense* and *SNUAlertApp* have been developed for sending and receiving emergency notifications. *SNUSense* is required to be installed on the accident victim's smartphone to collect and process the sensor's data. All the preprocessing methods, such as ALA calculation, complementary filter (sensor fusion), moving-maximum function, and machine learning classifiers (feature fusion) are programmed in the *SNUSense* application. *SNUAlertApp* is

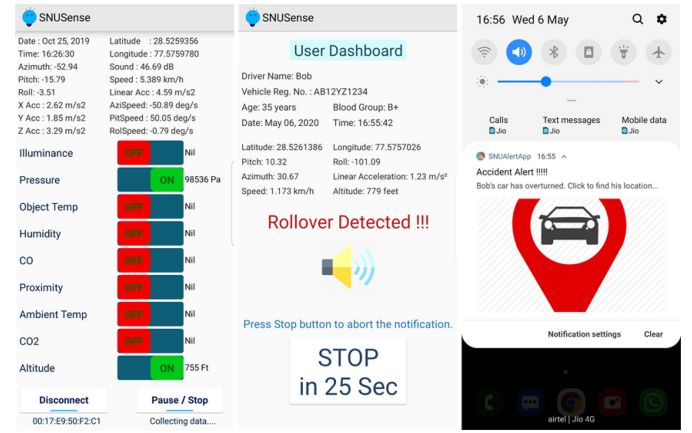


Fig. 5. (Left) data collection; (center) user dashboard; (right) notification.

required to be installed on accident rescuers' Android smartphone so that timely rescue operation can be initiated after locating the place.

1) *SNUSense Android Application*: An Android application, namely, *SNUSense*, has been developed for capturing and analyzing various parameters, such as rotation angle (roll, pitch, and azimuth), speed of the vehicle, noise, deceleration at different axes, atmospheric pressure, altitude, temperature, latitude, longitude, humidity, CO<sub>2</sub>, CO, light intensity, etc., by utilizing Android smartphone sensors and *Sensordrone* [Fig. 5(left)]. It continuously processes the streamed data coming from the *Sensordrone* and smartphone sensors and makes an estimate using the vehicle's five characteristics, that is, speed, ALA, change-in-altitude, pitch, and roll. After a positive detection of the accident event, *SNUSense* sends the name of the victim, type, and location of the vehicle accident to the *SNUSense* IoT server (*Google Firebase* [36]) using 4G Internet connection. We can see in the user dashboard that in case of false alarm, victim can abort the notification by pressing the STOP button [Fig. 5(center)]. IoT server further sends the notification (using *Firebase Cloud Messaging* service) to the intended recipients, such as EMS, police, family, etc., after analyzing the data (such as the severity of the accident based on the force of impact and the speed). It also collects all sensors' data in a CSV (comma-separated values) file in the smartphone's storage.

2) *SNUAlertApp Android Application*: Another Android application, *SNUAlertApp*, has been created to receive accident alerts as shown in Fig. 5(right). The alert notification contains the coordinates of the accident, the name of the driver, and the type and severity of the accident. A simple touch on the notification launches the *Google Maps* with a tracker icon on it so that rescuers can conveniently locate the place of accident.

## V. ACCIDENT SCENARIOS

Depending on the five variables described in Section III-C, this research focuses only on the identification and classification of four different vehicle accident situations: 1) collision; 2) falloff; 3) rollover; and 4) no accident. Table II describes the actions and criteria of five factors for each accident situation to be classified.

TABLE II  
ADC CRITERIA

| Parameter (Threshold) | ALA (5 g)                               | Speed (2 km/h) | Pitch / Roll (90°) | Altitude Change in 1 sec (8 feet) |
|-----------------------|-----------------------------------------|----------------|--------------------|-----------------------------------|
| <b>Collision</b>      | > 5 g                                   | < 2 km/h       | anything           | < 8 feet                          |
| <b>Fall-off</b>       | > 5 g                                   | < 2 km/h       | anything           | > 8 feet                          |
| <b>Rollover</b>       | Anything                                | < 2 km/h       | $\geq 90^\circ$    | anything                          |
| <b>No accident</b>    | Combinations other than above mentioned |                |                    |                                   |

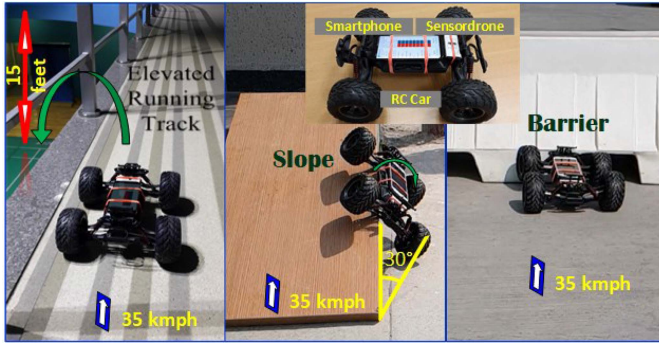


Fig. 6. (Left) falloff setup; (center) rollover setup; (right) collision setup.

Thresholds of speed and roll/pitch are identified by performing accident tests multiple times with our RC car setup. If any fatal accident occurs, the ideal speed of vehicle becomes zero instantly. Threshold for speed is considered 2 km/h because GPS readings does not stabilize immediately and slightly fluctuates around for a while. In the literature, for rollover, different researchers have considered different angles of roll/pitch to define the rollover. In this work, we are considering the rollover for angles greater than or equal to  $90^\circ$ , as we are considering not partial but complete rollover events. For collision, it is mentioned in the literature that if a vehicle collides with a solid obstacle with a speed of more than 23 km/h (i.e., 21 ft/s) then it will generate a deceleration of more than 5g and the collision is said to be severe [29]. From the following equation of velocity, we can observe that if a vehicle falls from 8 ft with an initial speed of 0, it achieves the speed of more than 21 ft/s in almost 0.8 s, which is minimum speed to generate 5-g deceleration:

$$\text{velocity}_{\text{final}} = \text{velocity}_{\text{initial}} + g * t. \quad (6)$$

So, threshold of parameter, i.e., *change-in-altitude*, is considered a minimum 8 ft. Although there may be other subclassification of the accidents, this study is restricted to collision, falloff, and rollover events. During the implementation of the classification, the lag between the reading timings of different parameters is well thought out.

## VI. EXPERIMENTAL SETUP AND DATA COLLECTION

### A. Experimental Setup

To accurately detect collision and falloff, the smartphone is placed near the center of gravity (CG) of the vehicle (in our case, on the armrest between the front seats) as both collision and falloff events leads to deceleration, and it is measured with the help of smartphone's built-in accelerometer sensor. Although the direction of impact on the vehicle

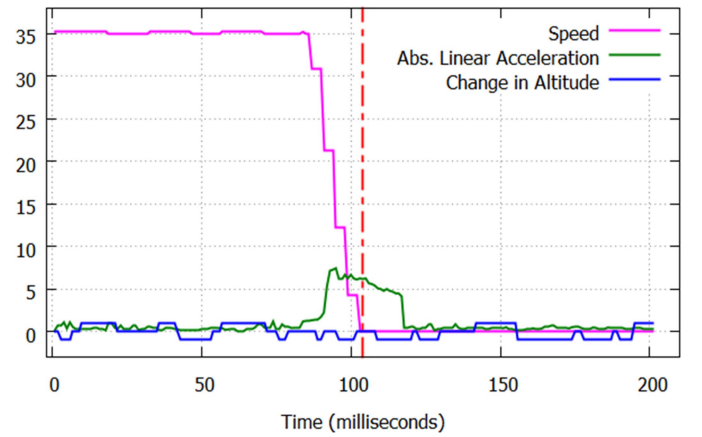


Fig. 7. Collision.

cannot be foreseen, the deceleration measurement would be more accurate if the direction of impact is same as the direction of the accelerometer sensor installation. Similarly, if the impact is on the opposite side of the sensor position, due to the size and build of the vehicle, the deceleration output may be less than the actual. The National Highway Traffic Safety Administration (NHTSA), the nodal agency of the U.S. Department of Transportation, recommends that inertial sensors be placed near the CG [37], for all crash tests. As shown in Fig. 6, we have also followed this recommendation for our experimental RC car setup and tied up the smartphone on the chassis of the RC car. For the rest of the sensors used in our experiment, their placement within a vehicle does not have a significant impact on the accuracy of their data values.

All the tests were performed at the speed of 35.1 km/h because, in the literature, minimum threshold speed to detect collision (or airbag deployment) of vehicle is  $\approx 23$  km/h [29]. It is considered that speed below 23 km/h will not be enough to create a severe collision impact. As speed will increase, the severity of accident will increase proportionally. Because this research does not focus on severity monitoring, considering the speed similar to the minimum speed would be sufficient to perform the accident tests. Another reason for considering experimental speed 35.1 km/h is the RC car's limitations. Although RC car's maximum speed is 53 km/h (which cannot be set to a specific speed), but due to the weights of smartphone and Sensordrone, it comes down to 35.1 km/h.

### B. Data Collection

To collect the head-on collision data, we have used a plastic road barricade as a barrier [Fig. 6(right)], and the RC car setup is crashed into it. Fig. 7 is showing the change in ALA, speed, and change-in-altitude during collision event. After the collision, a sudden decrease in the speed and increase in the ALA is clearly noticeable in the graph. The drawn red line reflects the moment when the collision occurred.

The falloff experiment is conducted on the elevated running track of the Indoor Sports Complex (ISC) at Shiv Nadar University, India [Fig. 6(left)]. The RC car setup was dropped onto the wooden tennis court from the running track elevated

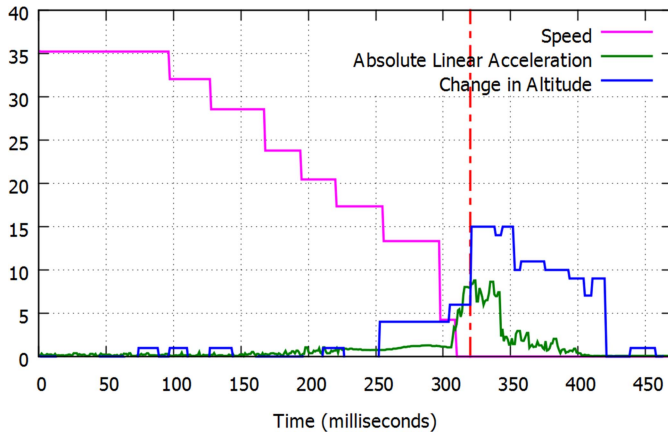


Fig. 8. Falloff.

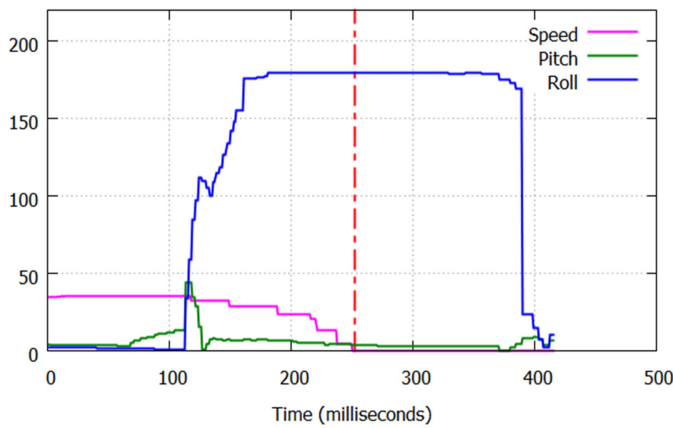


Fig. 9. Rollover.

by 15-ft height by diverting it to the left side with a 45° angle on the track. Fig. 8 reflects the falloff case, with changes in ALA, speed, and altitude. The gradual decrease in speed is showing the duration of fall, 15 ft change-in-altitude (the difference between the altitude before the fall and after the fall), and sudden increase in ALA is showing the percussion on the tennis court after the fall. The dotted red line indicates the moment of the percussion. 1-s sliding window has been used to record the change-in-altitude because in 1 s the falling object can attain the final speed of more than 30 km/h, which is far enough to create the impact force of more than 5 g.

To conduct the rollover experiments, we have used a 4-cm thick wooden board that was inclined 30° with the flat ground surface [Fig. 6(center)]. Fig. 9 demonstrates changes in speed, roll, and pitch angles during the rollover event. The sudden increase in the roll angle is the perfect indicator of rollover incident. The red dotted line in the graph shows the instant of the rollover activity.

Every sort of accident experiment is repeated 30 times to collect the data and 1167 observations has been recorded. Each plot in Figs. 7–9 are plotted using raw data samples received from the sensors, which are showing multivariate time series to observe the trends of different model parameters during different type of accidents.

## VII. ACCIDENT DETECTION AND CLASSIFICATION MODELS

In this work, three different classification models, namely, GMM (a clustering model), DT (a logic-based model), and NB (a probabilistic model) have been compared to find best learning model to address the vehicle's ADC problem. Five parameters of training data set, that is, change-in-altitude, pitch, roll, speed, and ALA has been used to train every classification model. Separate testing data set of each accident event has been used to assess the ADC performance of the models. 90% observations are used for training the models and 10% observations are used for testing the models.

### A. Gaussian Mixture Model-Based ADC Model

GMMs are among the most statistically matured methods for clustering and density estimation [38], [39]. They model the probability density function (PDF) of observed data points using a multivariate Gaussian mixture density. Mixture models are a form of density model that consists of several components, usually Gaussian in nature. These functions of components are combined to give a multimodal density. In this work, GMMs are developed to capture the information about each accident type, i.e., collision, rollover, and falloff.

The number of Gaussians in the mixture model is known as the number of components. They indicate the number of clusters in which data points are to be distributed. The number of components in each GMM is optimized based on the number of training data points. The components within each GMM capture finer level details among the feature vectors of each type of accident. Depending on the number of data points, the number of components may be varied in each GMM. If the GMM has a few components and is trained using a large number of data points, it may lead to more generalized clusters that fail to capture specific details related to each class. On the other hand, overfitting of the data points may happen, if too many components represent a few data points. Also, the complexity of the models increases if they contain a higher number of components.

The decision regarding the category of accident is taken based on its probability of coming from feature vectors of the specific model. Given a set of inputs, GMM refines the weights of each distribution through the expectation–maximization algorithm. Once a model is generated, conditional probabilities can be computed for test patterns (unknown data points). The ADC events are modeled using Gaussian PDFs, explained by the mean vector and the covariance matrix. For a feature vector  $x_t$ , the mixture density for an accident type is defined as the weighted sum of  $N$  component's Gaussian densities as

$$P(x_t|\Omega) = \sum_{i=1}^N W_i P_i(x_t) \quad (7)$$

where  $P_i(x_t)$  is the component densities and  $w_i$  are the weights. For a  $D$ -dimensional feature vector, each component density is a  $D$ -variate Gaussian function, i.e.,

$$P_i(x_t) = \frac{1}{(2\pi)^{\frac{D}{2}} |R_i|^{\frac{1}{2}}} e^{-\frac{1}{2}[(x_t - m_i)' \Sigma_i^{-1} (x_t - m_i)]} \quad (8)$$



the value of  $T$  based on  $X$ , we need to calculate the conditional probability of  $T$  given  $X$

$$p(T|X) = \frac{p(X|T)p(T)}{p(X)} \quad (13)$$

where,  $p(T)$  and  $p(X)$  are the constants that are directly derived from the data. To obtain the value of  $p(X|T)$ , it is factorized as

$$p(X|T) = p(x_1, x_2, \dots, x_n|T) = \prod_{i=1}^n p(x_i|T). \quad (14)$$

Combining (13) and (14), we get

$$p(T|X) = \frac{p(T)}{p(X)} \prod_{i=1}^n p(x_i|T). \quad (15)$$

The conditional distribution of  $T$  given  $X$  is calculated from (15). The value of target variable  $p(T|X)$  is classification outcome, which is the state of  $T$  with highest probability.

### VIII. RESULTS AND ANALYSIS

The performance of the ADC models based on GMM, NB, and DT classifiers are discussed in this section. Outcomes have been obtained using the SNUSense database. All 1167 experimental observations of the SNUSense database are shuffled using the random function, out of them 1050 (90%) observations are used for training the ADC models while 117 (10%) are used for testing. By an observation here we mean a data point that is the vector of values read from the sensors at a particular time during accident tests. The data set is balanced and large enough to train and test the model. The classification performance of the proposed models is assessed using precision, recall, F1-score, and receiver operating character (ROC) curve [44]. These quantities are described as following.

- 1) *Precision*: It is the ratio of the number of true positive (TP) predictions to all positive predictions. In this work, it represents the proportion of accident classifications that are actually right, i.e.,

$$\begin{aligned} \text{precision} &= \frac{\text{true\_positive\_samples}}{\text{positively\_predicted\_samples}} \\ &= \frac{\text{TP}}{(\text{TP} + \text{FP})}. \end{aligned}$$

- 2) *Recall*: It is the ratio of number of TPs to all the actual positive records. It represents the proportion of accident classifications that can be predicted by model, i.e.,

$$\text{recall} = \frac{\text{true\_positive\_samples}}{\text{actual\_positive\_samples}} = \frac{\text{TP}}{(\text{TP} + \text{FN})}.$$

- 3) *F1-Score*: It takes both precision and recall into consideration. It is the harmonic mean of precision and recall, i.e.,

$$\text{F1 - score} = 2 * \left( \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}} \right).$$

- 4) *ROC Curve*: The probability of detection, i.e., TP rate (TPR) is plotted as a function of the probability of false alarm, i.e., false-positive rate (FPR) in the

TABLE III  
DETECTION AND CLASSIFICATION PERFORMANCE USING NB

| Accident Type | TP | FP | FN | TN | Precision | Recall | F1-score |
|---------------|----|----|----|----|-----------|--------|----------|
| Collision     | 32 | 1  | 1  | 83 | 0.96      | 0.98   | 0.97     |
| Fall-off      | 27 | 2  | 2  | 86 | 0.92      | 0.94   | 0.93     |
| Rollover      | 30 | 1  | 2  | 84 | 0.96      | 0.95   | 0.96     |
| No-Acc.       | 22 | 1  | 1  | 93 | 0.94      | 0.95   | 0.94     |

TABLE IV  
DETECTION AND CLASSIFICATION PERFORMANCE USING GMM

| Accident Type | TP | FP | FN | TN | Precision | Recall | F1-score |
|---------------|----|----|----|----|-----------|--------|----------|
| Collision     | 31 | 3  | 2  | 81 | 0.91      | 0.94   | 0.92     |
| Fall-off      | 26 | 3  | 3  | 85 | 0.90      | 0.91   | 0.90     |
| Rollover      | 28 | 4  | 4  | 81 | 0.88      | 0.87   | 0.87     |
| No-Acc.       | 22 | 1  | 1  | 93 | 0.94      | 0.94   | 0.94     |

ROC curve for various threshold settings of the parameter. TPR [=TP/(TP + FN)] measures the proportion of actual positive incidents that are correctly identified. The FPR [=FP/(FP + TN)] is measured as the ratio between the number of falsely classified negative incidents as positive (false positives) and the overall number of TN incidents (irrespective of classification). TPR and FPR, both are different criteria which are compared in ROC curve. In our case, the area under the curve is an indicator of how well a function can distinguish a vehicle accident type with the rest of the others.

The detection and classification performance of the NB technique is summarized in Table III. The mean of precision, recall, and F1-score for the NB-based classification model are 0.94, 0.95, and 0.95, respectively. The model performs best with collisions, where the F1-score is 0.97, which is followed by a rollover, no accident, and falloff events.

Accident classification performance of GMM models for the four accident classes using the SNUSense database has been computed. Using the 1050 observations and 16, 32, 64, and 128 components for building each GMM, performances are observed to be 89.75%, 87%, 95.5%, and 90.5%, respectively. During the training period, it is observed that the highest accident classification performance of 95.50% is achieved with 64 components and 100 iterations. If more than 64 components are used, the performance deteriorates.

Table IV is a summary of the ADC performance of the GMM technique on testing data. The average precision, recall, and F1-score for the GMM-based model are observed to be 0.91, 0.92, and 0.91, respectively. In the accident events, the model performs best with collisions with an F1-score of 0.92, followed by falloff and rollover. The nonoccurrence of a road accident is reported with an F1 value of 0.94.

Table V describes the detection and classification performance of the DT model. The model performs best with rollover events with an F1-score of 0.92, followed by falloff and collision. The mean precision, recall, and F1-score of the DT model are observed to be 0.88, 0.89, and 0.88, respectively.

It is evident that the NB model is the best performing model with an average F1-score of 0.95, which is followed by the GMM and DT models with an F1-score of 0.91 and 0.88, respectively.

TABLE V  
DETECTION AND CLASSIFICATION PERFORMANCE USING DT

| Accident Type | TP | FP | FN | TN | Precision | Recall | F1-score |
|---------------|----|----|----|----|-----------|--------|----------|
| Collision     | 29 | 4  | 4  | 80 | 0.86      | 0.87   | 0.86     |
| Fall-off      | 26 | 3  | 3  | 85 | 0.90      | 0.89   | 0.89     |
| Rollover      | 30 | 3  | 2  | 82 | 0.92      | 0.93   | 0.92     |
| No-Acc.       | 20 | 3  | 3  | 91 | 0.86      | 0.88   | 0.87     |

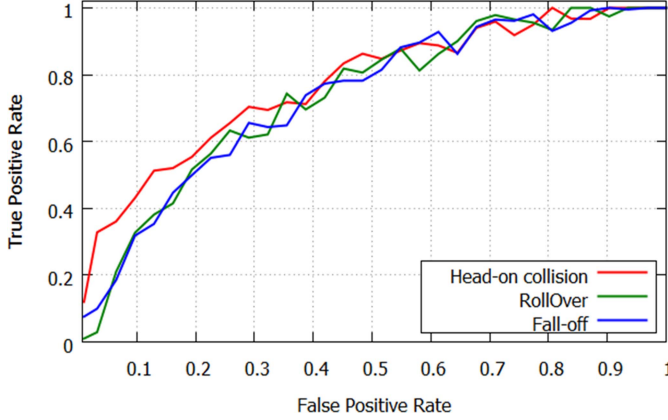


Fig. 12. ROC curve for NB.

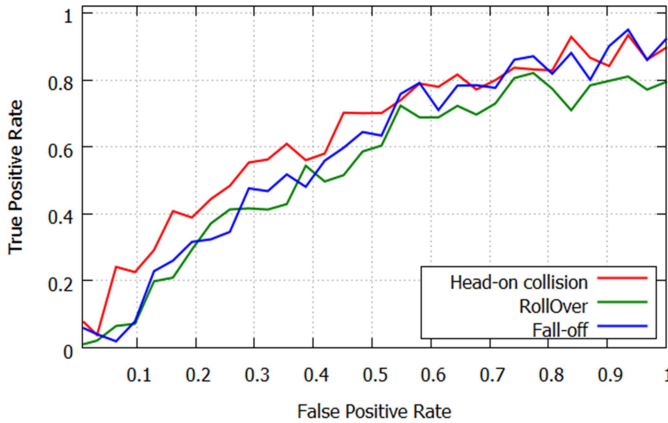


Fig. 13. ROC curve for GMM.

ROC curves of NB-, GMM-, and DT-based models are plotted in Figs. 12–14, respectively. Actually, the ROC curve is a graphical plot that demonstrates a binary classifier system's diagnostic capability. Because our problem is a multiclassification problem, each ROC curve of a particular model is drawn by using the One-versus-Rest scheme where each accident scenario is plotted against all other accident types, which makes it binary classification problem for ROC curves of every accident types. With the help of positive and negative outcomes of ADC models, we have used binary logistic regression and saved the prediction probabilities. Then for generating the multivariable ROC curve for a particular accident class, saved prediction probabilities as the test variables are utilized.

To train, test, and identify the best ADC model among NB-, GMM- and DT-based models, first the “*scikit-learn*” Python library and SNUSense data set are used on a separate computer. After identifying that the NB-based model is the best ML model over GMM and DT for our ADC problem, the

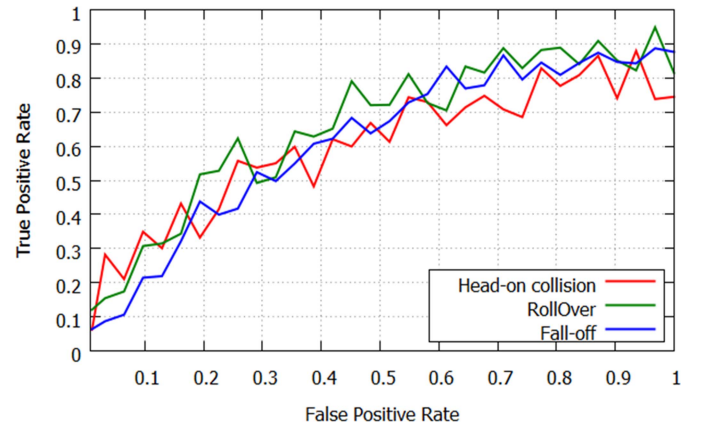


Fig. 14. ROC curve for DT.

NB-based model is implemented in the SNUSense application. We have used the “JNBC” Java library [45] for NB classification, which is provided by NamSor.com. Although, every ML algorithm is hard to implement in smartphones because of their processing toll and availability of a suitable library. For this purpose, the concept of IoT server is used in the proposed architecture to implement the ML algorithm on the server side and to establish publish/subscribe communication.

In case of rollover and falloff accidents, RC cars can mimic the real car accidents completely. As far as the intensity of deceleration generated due to the impact is concerned, it may differ due to the build, size, and weight of a real car, thus deceleration threshold could be set accordingly while training the system for real cars.

## IX. CONCLUSION

The proposed detection system works frequently whether there is an accident or not and reports the incident to predefined emergency services and family in case of an accident. The system classifies accidents into four classes, i.e., collisions, rollovers, fallofts, and no accidents, so that the best possible rescue operations can be undertaken. Five training variables namely, change-in-altitude, pitch, roll, speed, and ALA are used as input variables to train and test the system. The proposed ADC system uses smartphone sensors, and Sensordrone sensors to measure the values of model variables. The system can be retrofitted in any type of vehicle.

As far as the accuracy of the proposed ADC system is concerned, three different GMM-, NB-, and DT-based classification models are evaluated and compared to determine the most accurate ADC model. The NB model is found to outperform others with an average F1-score of 0.95. It is the most accurate for collision with an F1-score of 0.97, followed by rollover, no accident, and falloft events, respectively.

If someone wants to reduce the time of automatic notification after the incident or compare the system with other automatic notification systems, every delay should be as short as possible whether it is algorithm's execution time or time taken in notification. But this work focuses on reducing overall reporting time using technology, compared to just manual reporting or nonreporting of an incident. To achieve this, we

have tried to reduce the time taken for notification after accident detection by implementing machine learning algorithms on the smartphone itself rather than transmitting data of all sensors after every 10 ms.

The system is low cost in comparison to factory fitted systems because they substantially increase the costs of the cars. According to [46], there were more than 3.2 billion smartphone users in the world in 2019. Hence, we have used a smartphone and its built-in sensors to develop our system. Apart from the smartphone, only a barometric altimeter and a 4G Internet connection are needed to implement the system.

To the best of our knowledge, our system is the only one-of-a-kind system that classifies accidents as collision, rollover, falloff, and no accident. Although system is highly accurate and have several advantages over other systems, it also has some limitations, i.e.: 1) system needs continuous Internet connection to send the emergency alerts; 2) placement of the smartphone would be predefined and user cannot put his smartphone at any other place, such as pocket, bag, etc., as otherwise chances of FP (in case of no accident) and FN (in case of an accident) would be very high; and 3) if smartphone gets ejected outside the vehicle, or hardware setup breakdowns, results may be impacted, and system may fail.

As a future work, we intend to add another accident class, i.e., fire/explosion, to our ADC model. To identify the most appropriate classification model for our ADC system, we are going to implement some validation schemes, such as K-fold cross validation, and we will also consider the time taken by the classification algorithm. In this work, both Android apps are function oriented, but in future we are also going to merge them in a single app having a user-friendly and interactive interface to make a unified solution for our ADC system.

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